

Data Analytics ISE Activity: Datathon

Project Title: Customer Shopping Behavior Analysis

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Date: 6th April 2026

1. Project Overview

This project focuses on analyzing customer shopping behavior using data analysis techniques in Python. The dataset contains customer demographics, purchase details, and shopping patterns. The goal is to clean, preprocess, and transform the data to extract meaningful insights that can help businesses understand customer preferences and improve decision-making.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features: - Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type) -
Missing Data: 37 values in Review Rating column

3. Objectives

- Analyze customer purchasing behavior
- Understand customer segmentation based on age
- Study purchase frequency patterns
- Prepare data for further analysis or machine learning
- Store processed data for database use

4. Dataset Description

The dataset includes customer-related information such as:

- **Age** – Customer age
- **Category** – Product category
- **Purchase Amount (USD)** – Amount spent
- **Frequency of Purchases** – Shopping frequency
- **Review Rating** – Customer rating
- **Discount Applied / Promo Code Used** – Promotional activity

5. Data Preprocessing

We began with data preparation and cleaning in Python:

A. Data Loading: Imported the dataset using pandas.

```
[2]: # Loading the dataset using pandas
import pandas as pd
df = pd.read_csv('customer_shopping_behavior.csv')
```

B. Data Overview

- Initial Exploration: Used `df.info()` to check structure and `.describe()` for summary statistics.

```
[6]: df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  ---                                ---
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   object
3   Item Purchased                      3900 non-null   object
4   Category                            3900 non-null   object
5   Purchase Amount (USD)               3900 non-null   int64
6   Location                            3900 non-null   object
7   Size                                3900 non-null   object
8   Color                               3900 non-null   object
9   Season                              3900 non-null   object
10  Review Rating                       3863 non-null   float64
11  Subscription Status                 3900 non-null   object
12  Shipping Type                      3900 non-null   object
13  Discount Applied                   3900 non-null   object
14  Promo Code Used                    3900 non-null   object
15  Previous Purchases                  3900 non-null   int64
16  Payment Method                     3900 non-null   object
17  Frequency of Purchases              3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

[10]:	# Summary Statistics using .describe() df.describe(include = 'all')														
[10]:		Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
	count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
	unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2
	top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No
	freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223
	mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
	std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
	min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
	25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
	50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
	75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
	max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

C. Handling Missing Values

Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

D. Column Renaming

- Improves readability
- Makes column names consistent

6. Feature Engineering

- Age Group Creation:- Customers are divided into:
 - Young Adult
 - Adult
 - Middle-aged
 - Senior

This helps in **customer segmentation**

- Purchase Frequency Conversion

- Converts categorical frequency into numeric values (days)
- Makes it usable for analysis & ML

7. Exploratory Data Analysis (EDA)

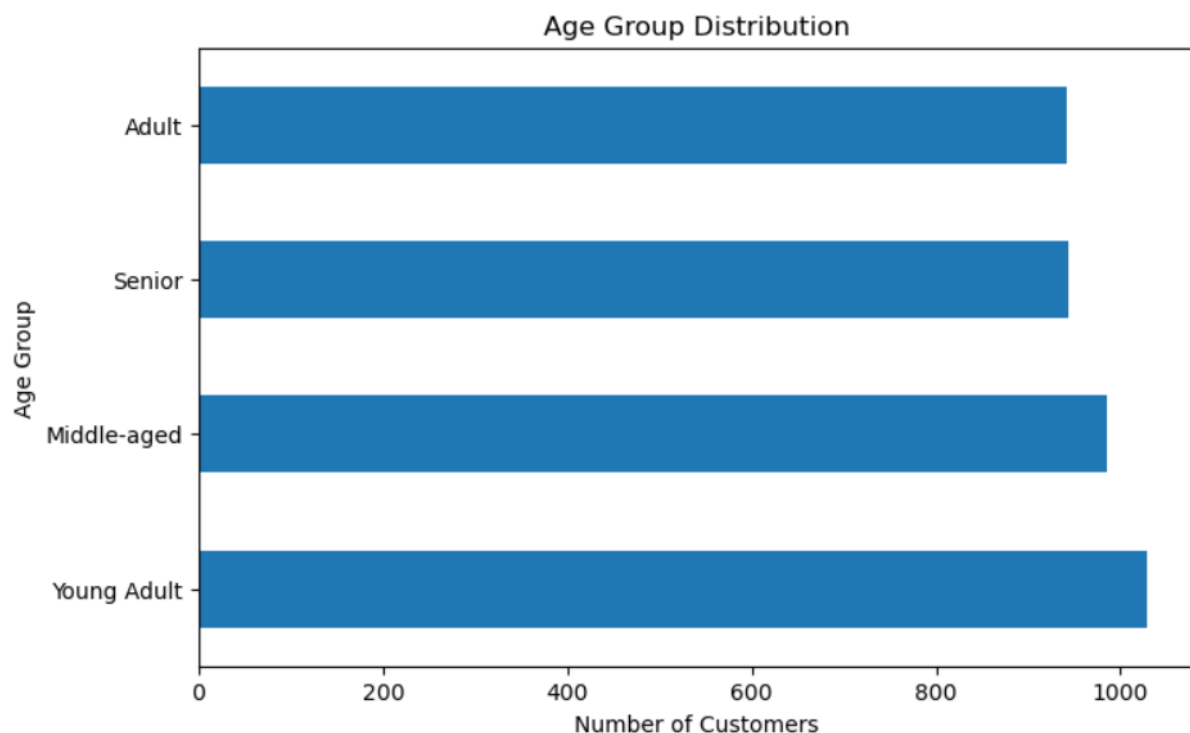
Example 1: Age Group Distribution (Bar Chart)

Explanation:

- Shows how customers are distributed across age groups
- Helps identify dominant customer segment

Observation:

Most customers belong to the Adult and Middle-aged group, indicating higher purchasing activity in these segments.



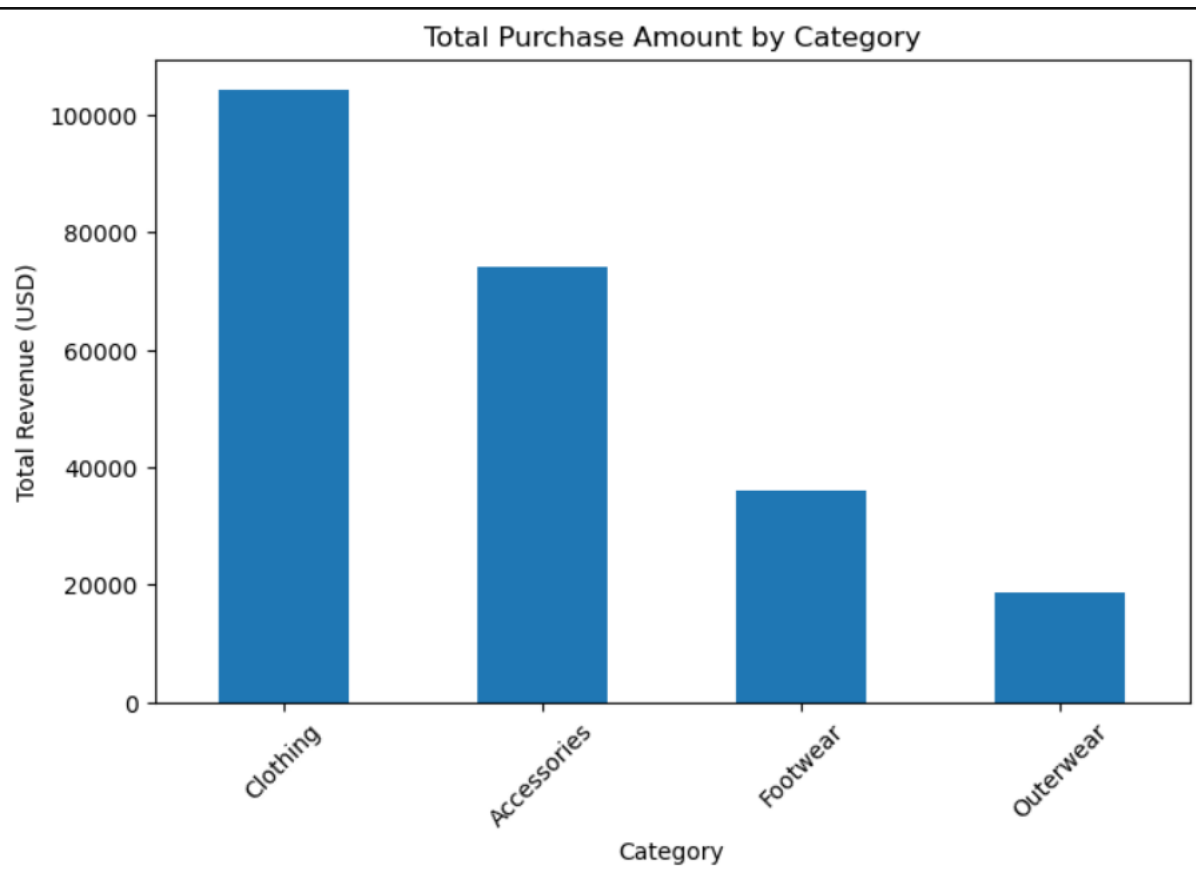
Example 2: Purchase Amount by Category

Explanation:

- Compares spending across product categories

Observation:

Certain categories(clothing) generate higher revenue, indicating customer preference and demand.



8. Database Integration

Performed structured analysis in MYSQL to answer key business questions:

```
Local instance MySQL90 x
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
  Filter objects
  activity
  customer_behavior
    Tables
      customer
    Views
    Stored Procedures
    Functions
  practical2
  practical3
  practical4
  practical5
  practical6
  practical7
  sakila
  sys
  world
Administration Schemas
Information
No object selected

Query 1 query*
Limit to 1000 rows

7
8 -- Q2. Which customers used a discount but still spent more than the average purchase amount?
9 • select customer_id,purchase_amount
10 from customer
11 where discount_applied = 'Yes' and purchase_amount >=(select AVG(purchase_amount) from customer);
12
13 -- Q3. Which are the top 5 products with the highest average review rating?
14 • SELECT
15     item_purchased,
16     ROUND(AVG(CAST(review_rating AS DECIMAL(10,2))), 2) AS Avg_product_rating
17 FROM customer
18 GROUP BY item_purchased
19 ORDER BY avg_product_rating DESC
20 LIMIT 5;
21
22
23 -- Q4. Compare the average Purchase Amounts between Standard and Express Shipping.
24 • select shipping_type,
25     ROUND(AVG(purchase_amount),2)
26 from customer
27 where shipping_type in ('Standard','Express')
28 group by shipping_type;
29
30 -- Q5. Do subscribed customers spend more? Compare average spend and total revenue
31 -- between subscribers and non-subscribers.
32 • SELECT subscription_status
```

9. Dashboard in Power BI



10. Conclusion

The project successfully preprocesses and structures customer shopping data. By transforming raw data into meaningful formats, businesses can better understand customer behavior, optimize marketing strategies, and improve decision-making.